MCC Entrance Test, August 2026 — Day 2

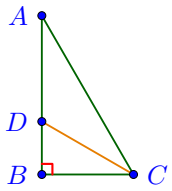
Level T2

Test Information. Level T2 — 120 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Algebra, Combinatorics, Geometry (×2), Number Theory (×2).
Rules: show your work (partial credit rewards genuine progress); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

Problem 1 (10 pts). Two numbers x and y satisfy $x + y = 4$ and $xy = 3$. What is the value of $(x - y)^2$?

Problem 2 (10 pts). A rectangular box (a cuboid) has length 3, width 3, and height 23. A cube has the same total surface area as this box. Find the side length of the cube.

Problem 3 (15 pts). In right triangle ABC, the right angle is at B. Point D lies on segment AB with $DB = 10$. Given that $\angle CDB = 60^\circ$ and $\angle ACD = 30^\circ$,



(a) Find the length of CD. [6 pts]

(b) Find the length of AD. [9 pts]

Problem 4 (15 pts). A trench has two rows of pits, one row along each side, with 10 pits in each row. A total of 7 diggers are to be placed into pits so that, within each row, no two diggers are in adjacent pits (a digger in one row may be next to a digger in the other row — only pits in the *same* row count as adjacent).

(a) In one row of 10 pits, find the number of ways to place k diggers with no two adjacent, for each $k = 0, 1, 2, 3, 4, 5$. [9 pts]

(b) Using part (a), find the number of ways to place all 7 diggers across the two rows, following the rule above. [6 pts]

Problem 5 (25 pts). Consider the numbers $1, 2, 3, \ldots, 99$, and the alternating sum formed by adding and subtracting their squares in order, starting with a plus sign:

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots - 98^2 + 99^2 = \underbrace{(1^2 - 2^2)}_{k=1} + \underbrace{(3^2 - 4^2)}_{k=2} + \cdots + \underbrace{(97^2 - 98^2)}_{k=49} + 99^2.$$

(a) Show that the k-th paired term $(2k - 1)^2 - (2k)^2$ simplifies to $-(4k - 1)$. [9 pts]

(b) Using part (a), evaluate $\sum_{k=1}^{49} -(4k - 1)$, the sum of all 49 paired terms. [3 pts]

(c) Add back the unpaired term 99^2 to find the value of the full alternating sum, then find the greatest prime number that divides it. [13 pts]

Problem 6 (25 pts). A two-digit number is called *lucky* if it is evenly divisible by the sum of its own digits. (For example, 12 is lucky since $1 + 2 = 3$ divides 12; but 13 is not, since $1 + 3 = 4$ does not divide 13.) How many lucky two-digit numbers are there?

(a) Write the number as $\overline{tu} = 10t + u$, with tens digit $t \in \{1, \ldots, 9\}$ and units digit $u \in \{0, \ldots, 9\}$. Show that $(t + u) \mid (10t + u)$ if and only if $(t + u) \mid 9t$. [7 pts]

(b) Explain why, for a fixed tens digit t, this restricts $t + u$ to being one of the (finitely many) divisors of $9t$ that also lie in the range $[t, t + 9]$ (the range $t + u$ can take as u runs over $0, \ldots, 9$). [3 pts]

(c) Using parts (a)–(b), check each tens digit $t = 1, \ldots, 9$ and count the total number of lucky two-digit numbers. [15 pts]